

Attachment 2 — Roadmap with hour estimates

NLnet NGI Zero Commons Fund — Symphact application, 13th open call (deadline 2026-06-01).
Condensed from docs/roadmap-en.md v1.0. Full text: roadmap-en.md .

Summary table

Phase	Milestones	Est. hours	Delivers
Phase 1 Core Runtime	M0.1–M1.0	~148–204	Actor runtime + CFPU reference impl
Phase 2 Kernel Foundation	M2.1–M2.5	~120–160	Scheduler, router, memory, capabilities
Phase 3 Device & I/O	M3.1–M3.5	~80–120	Device actors (UART, GPIO, timer, storage)
Phase 4 Advanced Runtime	M4.1–M4.4	~100–140	Hot code loading, migration, backpressure
Phase 5 Security & Compliance	M5.1–M5.4	~100–140	Capability security, audit, formal verification
Phase 6 Developer Experience	M6.1–M6.5	~100–140	CLI, IDE, profiler, NuGet, docs
Phase 7 First Sellable Product	M7.1–M7.5	~80–120	Reference apps, enterprise, launch
Total		~728–1024	Production-ready Symphact

Estimates are calibrated to M0.2 actuals (~8 hours, 580 lines of code, 46 tests, medium complexity).

Phase 1 — Core Runtime (M0.1–M1.0)

Milestone	Status	Tests	Est. hours	Highlights
M0.1 Actor core primitives	✓	30	~6	<code>IMailbox</code> / <code>TMailbox</code> , <code>TActorRef</code> , <code>TActor<TState></code> , <code>TActorSystem</code>
M0.2 ActorContext + inter-actor messaging	✓	+16	~8 (actual)	<code>IActorContext</code> / <code>TActorContext</code> , <code>DrainAsync</code> with <code>MaxRounds</code>
M0.3 Supervision (let-it-crash)	✓	+30	~24–32	<code>ISupervisorStrategy</code> , <code>OneForOne/AllForOne</code> , lifecycle hooks
M0.4 Scheduler + per-actor parallelism	✓	+86	~24–32 (actual ~6h agent team)	<code>IScheduler</code> , <code>TInlineScheduler</code> , <code>TDedicatedThreadScheduler</code> , <code>QuiesceAsync</code>
M0.5 Persistence — BCL-only slices	✓	+44	shipped	<code>IJournal</code> + <code>TInMemoryJournal</code> , <code>ISnapshotStore</code> + <code>TInMemorySnapshotStore</code>
M0.5 Persistence — content-addressed	⚠	—	~16–24	<code>TCasJournal</code> + <code>TCasSnapshotStore</code> (SHA-256), supervision lifecycle integration
M0.6 Remoting / Distribution	⌚	—	~28–36	<code>ITransport</code> + <code>TTcpTransport</code> , serialization, location-transparent <code>Send</code>
M0.7 CFPU Hardware Integration	⌚	—	~30–50	<code>TMmioMailbox</code> , end-to-end demo over <code>FenySoft.CilCpu.Sim</code>
M1.0 Samples + Stabilization	⚠ (CounterActor ✓ , többi tervezett)	sample futtatható	~12–16	<code>CounterActor</code> (✓) leszállítva mint runnable sample, <code>samples/CounterActor/</code> , <code>ChatRoom</code> , API reference, benchmarks

Total green tests at grant submission: 186 (120 Core + 22 Platform.DotNet + 44 Persistence).

Phase 2 — Kernel Foundation (M2.1–M2.5, ~120–160h)

Milestone	Est. hours	Description
M2.1 Root Supervisor + Actor Hierarchy	~20–28	<code>TRootSupervisor</code> , boot sequence, fixed restart strategy
M2.2 Scheduler Actor	~28–36	<code>ISchedulerPolicy</code> , <code>TRoundRobinPolicy</code> , <code>TLoadBalancePolicy</code> , actor-to-core affinity, watchdog
M2.3 Router Actor	~24–32	<code>TRouter</code> , location-transparent <code>Send</code> , migration support, dead-letter handling
M2.4 Memory Manager	~20–28	<code>TMemoryManager</code> , per-core SRAM budget, per-actor GC, memory pressure notification
M2.5 Capability Registry	~28–36	<code>TCapabilityRegistry</code> , CST HW table management, delegation + revocation, 8-bit permission flag

Phase 3 — Device & I/O Layer (M3.1–M3.5, ~80–120h)

Milestone	Est. hours	Description
M3.1 Device Actor Framework	~20–28	<code>IDeviceActor</code> , MMIO ownership, interrupt → message conversion
M3.2 UART Device Actor	~12–16	TX/RX actor pair, subscribe pattern, backpressure
M3.3 GPIO Device Actor	~8–12	Pin config (Input/Output/Interrupt), edge-triggered IRQ → message
M3.4 Timer / Clock Service	~12–16	<code>TTimerService</code> , <code>AskWithTimeout</code> , mock clock for testing
M3.5 Storage Service	~28–36	<code>IStorageProvider</code> , <code>TFileSystemActor</code> , file handle = actor

Phase 4 — Advanced Runtime (M4.1–M4.4, ~100–140h)

Milestone	Est. hours	Description
M4.1 Hot Code Loading	~32–40	<code>THotCodeLoader</code> , opcode whitelist, Erlang-style version switch, state preservation
M4.2 Actor Migration	~24–32	Suspend → serialize → move; router mapping update; cross-core / cross-node
M4.3 Backpressure + Flow Control	~20–28	Mailbox depth limit, <code>TrySend</code> , backpressure propagation
M4.4 Priority Mailbox + System Messages	~24–32	<code>TPriorityMailbox</code> , opt-in per actor, watchdog kill, graceful shutdown

Phase 5 — Security & Compliance (M5.1–M5.4, ~100–140h)

Milestone	Est. hours	Description
M5.1 Full Capability Model	~28–36	CST HW lookup on every Send, audit trail, AuthCode integration
M5.2 Capability Delegation & Revocation	~20–28	Attenuation on delegation, revocation broadcast
M5.3 Audit & Compliance Logging	~20–28	Tamper-proof audit log actor, IEC 61508 / ISO 26262 baseline reporting
M5.4 Formal Verification Foundations	~32–48	TLA+ / Dafny spec of <code>send</code> / <code>receive</code> semantics, capability invariants

Phase 6 — Developer Experience (M6.1–M6.5, ~100–140h)

Milestone	Est. hours	Description
M6.1 CLI Tooling	~20–28	<code>symphact</code> CLI: <code>new</code> , <code>build</code> , <code>run</code> , <code>deploy</code> , <code>monitor</code>
M6.2 IDE Integration	~24–32	VS Code extension, Rider plugin, live actor tree, message breakpoint
M6.3 Profiler & Monitoring Dashboard	~24–32	Per-actor msg/sec, latency histogram, Prometheus + Grafana export
M6.4 NuGet Packages & API Reference	~16–24	<code>Symphact.Core</code> , <code>Symphact.Supervision</code> , <code>Symphact.Persistence</code> , <code>Symphact.Remoting</code> , <code>Symphact.Security</code> , <code>Symphact.Devices</code>
M6.5 Documentation & Tutorials	~16–24	Getting started, architecture deep-dive, Akka.NET / Orleans migration guide

Phase 7 — First Sellable Product (M7.1–M7.5, ~80–120h)

Milestone	Est. hours	Description
M7.1 Reference Applications	~24–32	IoT Gateway, AI Agent Cluster, SNN Demo, Chat Server
M7.2 Production Hardening	~20–28	10K+ actor stress test, chaos testing, soak test, performance regression CI
M7.3 Enterprise Features	~16–24	RBAC, multi-tenant isolation, configuration management, K8s health check
M7.4 Licensing & Packaging	~8–12	Dual license, Docker image, Helm chart, landing page
M7.5 Launch Preparation	~12–16	Announcement blog, conference talks, GitHub Sponsors, first partner onboarding

Grant-funded scope (12 months, €30 000)

The grant funds the **next stretch beyond what has shipped** — the remaining M0.5 work, M0.6, M0.7, plus selected pieces of Phase 5 (formal-verification foundations) and Phase 6 (NuGet, docs, outreach). Hot code loading (M4.1) is **explicitly deferred** to a follow-up grant.

Grant milestone	Roadmap mapping	Hours	Amount	Timeframe
M1: Persistence	M0.5 (content-addressed)	~80h	€2 900	Months 1-3
M2: Remoting + Capability Registry	M0.6 + M2.5	~160h	€5 800	Months 2-7
M3: Kernel + Device actors	M2.1, M2.3, M3.1-M3.4	~180h	€6 500	Months 5-10
M4: CFPU integration demo	M0.7 (partial)	~80h	€2 900	Months 8-11
M5: Dev experience + docs + outreach	M6.4 + M6.5 (partial)	~180h	€6 500	Continuous
M6: Formal-verification foundations	M5.4 (partial)	~80h	€2 900	Months 9-12
Engineering hours total		~760h	€27 500	12 months
Online outreach + documentation	(video production, GitHub Pages, doc expansion)	—	€2 500	Continuous
Requested amount			€30 000	

Expected `osreq-to-cfpu` issues: 3-7 concrete HW requirements documented over the 12 months — feedback paths include SHA-256/BLAKE3 hash instruction, CST cache, mailbox FIFO depth, supervision fault-notification primitives, SRAM-to-SRAM DMA.